Data Preparation

1. CSV files were downloaded from the Kaggle dataset: https://www.kaggle.com/datasets/olistbr/brazilian-ecommerce
2. Column names, data types and formatting were adjusted; unnecessary columns were removed.
3. Product category names were adjusted by collapsing near-synonymous values into unified categories.
4. Review processing
    a. Due to high processing power required for translation from Portuguese to English, previously translated files shared on Kaggle by the user @Slime were used: https://www.kaggle.com/datasets/slimeee8888/english-translated-reviews-for-brazilian-e-comm
    b. "review_comment_title" and "review_comment_body" columns were merged into a single "full_review" column
    c. Python scripts written with the assistance of Claude were used in Google Colab for sentiment analysis and phrase count:
        i. Sentiment analysis steps (script: https://github.com/joanna-wi/joanna-wi.github.io/blob/Project-1/Scripts/olist_sentiment_analysis.py)
            1. Data setup
            2. Sentiment scoring of each review
            3. Key phrase extraction for the purpose of subsequent concept formation
            4. Concept-level sentiment analysis
            5. Results export
        ii. Phrase count steps (script: https://github.com/joanna-wi/joanna-wi.github.io/blob/Project-1/Scripts/themes_summary.py)
            1. Data setup
            2. Key phrase extraction from each review
            3. Phrase clustering
            4. Results export
            5. Following script execution, near-synonymous phrases were grouped, spelling errors corrected, and Claude used to generate an image with 50 most frequent phrases, wherein each phrase was coloured using red-green scale to represent the corresponding average sentiment
    d. "concept_overall" file was imported into PowerBI as a source for concept sentiment analysis visual, whereas the word map was copied directly to the dashboard.

Dashboard Construction

1. Report-level filters
    a. Orders placed in September and October 2018 were filtered out due to missing data.
    b. Orders with blank values in "items_ordered" column were filtered out due to missing data.
2. Measures were created to derive:
    a. Customer count (overall, one-time, repeat)
    b. Order count
    c. Revenue
    d. Within-order item and category counts
    e. Shares of total orders/revenue
    f. Score averages
    g. Written review counts
    h. Dynamic titles

    i. “Selected metric” variables for visuals with slicers
3. Calculated tables were created to obtain:
    a. Slicers
    b. Within-order category pairings
    c. Input data for Key Influencers analysis of score
4. Start of month/quarter/year columns were created for display of temporal changes in revenue, order and customer volume. A parameter was implemented for switching between periods.
5. Key Influencers visual was implemented to determine factors impacting review ratings.
6. Map of client locations was constructed using coordinates from “olist_geolocation_dataset” source.
7. Buttons and bookmarks were implemented for report navigation.

Report Preparation

1. The report was built in PowerPoint.
2. Graphs were constructed in Excel using data from the dashboard.
3. The image on the title slide was downloaded from Pexels: https://www.pexels.com/photo/a-miniature-shopping-cart-and-bag-on-a-blue-background-6214383/
4. The version of the Olist logo displayed in the report was downloaded from Mercos: https://mercos.com/uploads/olist_logoolist_negativo_rastroamarelo_rgb.png
5. Icons were downloaded from Flaticon:
    a. Auto
    b. Up arrow
    c. Telephone
    d. Kitchenware
    e. Health beauty
    f. Furniture decor
    g. Computer accessories
    h. Watch gift
    i. Cool emoji
    j. Garden tools
    k. Morning routine
    l. Baby
    m. Construction tools
    n. Living room
    o. Cart
    p. Toys
    q. Home appliances
    r. Deadline
    s. Expansion
    t. Positive review
    u. Revenue increase
6. Map was constructed using MapChart: https://www.mapchart.net/brazil.html